

Air University



Organization & Assembly (Lab)

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Ms. Sonia Shehzadi

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Department of Cyber Security

USAMA ARSHAD | 247141

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Lab Task # 11

Conditional Logic in Assembly

Write an assembly program that reads a character from a user, if it is either 'y' or 'Y' display it, otherwise terminate the program.

Code

```
.MODEL SMALL
.STACK 100h

.DATA
    prompt      DB "Enter 'Y' or 'y' to display the character: $"
    NEWLINE_CR  EQU 13                ; Carriage Return
    ASCII
    NEWLINE_LF  EQU 10                ; Line Feed ASCII

.CODE
MAIN PROC
    ; Initialize data segment
    MOV AX, @DATA
    MOV DS, AX

    ; --- 1. Display Prompt ---
    MOV AH, 09h                ; DOS interrupt for displaying a string
    LEA DX, prompt              ; Load effective address of 'prompt'
    INT 21h

    ; --- 2. Read Character (with echo) ---
    MOV AH, 01h                ; Function 01h: Read character WITH echo (shows input immediately)
    INT 21h                    ; AL now contains the input character, and it has been displayed

    ; --- 3. Print a Newline (to move cursor before final output) ---
    ; Note: We save AL before using AH/DI
    PUSH AX                    ; Preserve the input character (in AL)

    MOV DL, NEWLINE_CR         ; Carriage return
    MOV AH, 02h                ; Function 02h: Display character in DL
    INT 21h

    MOV DL, NEWLINE_LF         ; Line feed
    MOV AH, 02h
    INT 21h

    POP AX                     ; Restore the input character back to AL
```

```
; --- 4. Conditional Logic Check ---

; Check if input is 'Y' (uppercase)
    CMP AL, 'Y'
    JE ShowChar          ; If equal, jump to ShowChar

; Check if input is 'y' (lowercase)
    CMP AL, 'y'
    JE ShowChar          ; If equal, jump to ShowChar

; If neither 'Y' nor 'y', jump to exit
    JMP EXIT

ShowChar:
; --- 5. Display the Character (on the new line) ---
    MOV DL, AL           ; Move the character from AL to DL for output
    MOV AH, 02h          ; DOS interrupt for writing a character
    INT 21h

EXIT:
; --- 6. Terminate the program ---
    MOV AH, 4Ch          ; DOS interrupt for program termination
    INT 21h

MAIN ENDP
END MAIN
```

Explanation:

- 1. Initialization:** The program first sets the DS (Data Segment) register to correctly locate variables like the prompt string in memory.
- 2. Input with Echo:** It uses DOS function AH=01h to read a single character from the keyboard. This function immediately **echos** the character, displaying the user's input, and stores the character in the AL register.
- 3. Cursor Reset:** A new line (CR and LF) is printed using AH=02h to move the cursor down. During this process, the input character in AL is temporarily saved and restored using PUSH AX and POP AX to prevent corruption.
- 4. Conditional Check:** The CMP instruction checks the character in AL against both 'Y' (uppercase) and 'y' (lowercase). If either condition is true, the JE (Jump if Equal) instruction redirects the program flow to the ShowChar section.
- 5. Output Logic:** If the input was 'Y' or 'y', the program executes the ShowChar block, moving the character from AL to DL and using AH=02h to display it on the new line. If the input was anything else, the JMP EXIT instruction is executed.
- 6. Termination:** The program ends cleanly by executing DOS function AH=4Ch, returning control to the operating system.

Output

Output when Y or y is entered:

```
D:\>D:\ntest
Enter 'Y' or 'y' to display the character: y
y
```

Output when any other character is entered:

```
D:\>D:\ntest
Enter 'Y' or 'y' to display the character: x
```

THE END
